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# Kestrel Talent

An autonomous hiring agent that buys information only when it pays for itself

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**#2 of 51**
final rank on the arena leaderboard

**1,491.2**
arena score: points − credits × 0.05

**116 / 116**
slots filled, every signing verified

**9,876**
credits used: 20% of the 50,000 budget

Grand finale · 28 September 2026 · IIT Delhi

# The challenge, and the one rule we built on

## The arena

**1,000,000 candidates, one shared pool**
51 agents hunt it at once. Once a rival signs a name, it is gone.

**10 requisitions, 116 slots**
Each has a bar: role, assessment, notice, CTC. A signing below the bar scores zero.

**50,000 credits, every call metered**
Arena score = points held at the whistle − credits used × 0.05.

**Recon → market → closing hour**
Offers are locked for the first hour and cost double in the last.

## What each action costs, in points

| Action | Credits | Points at 0.05 |
|---|---|---|
| Search: 100 summaries | 1 | **0.05** |
| One full profile | 2 | **0.10** |
| Batch: 50 full profiles | 60 | **3.00 (0.06 each)** |
| Verified assessment | 25 | **1.25** |
| Offer (closing hour) | 10 (20) | **0.50 (1.00)** |
| Release a signing | 5 | **0.25** |

**17.1** points earned by an average verified signing: information is cheap next to a good hire

**Our one rule: buy an action only when its expected points exceed its credits × penalty factor.**
PENALTY_FACTOR is an environment variable set at kick-off. Every threshold in the code is derived from it; none is hard-coded.

03 // RESULTS

# #2 of 51, on a fifth of the budget

**1,985**
points held at the whistle

**9,876**
credits: recon 6,475 · race 1,270 · swap pilot 2,131

**17.1**
points per signing, all 116 verified

**10 s**
to sign 115 of 116 slots once offers opened

## Final leaderboard · 51 teams

| | | |
|---|---|---|
| 1 | confused_devs | 1,827.3 |
| 2 | alpha_beta (us) | 1,491.2 |
| 3 | ctrl_alt_defeat | 1,462.2 |

Our score: 1,985 points − 9,876 credits × 0.05 = 1,491.2

## How the day went (IST)

**11:04** — Launch; recon starts buying

**11:29** — Verified queues full (6,225 credits)

**11:42** — Offers open 12 min late: 115 signed in 10 s

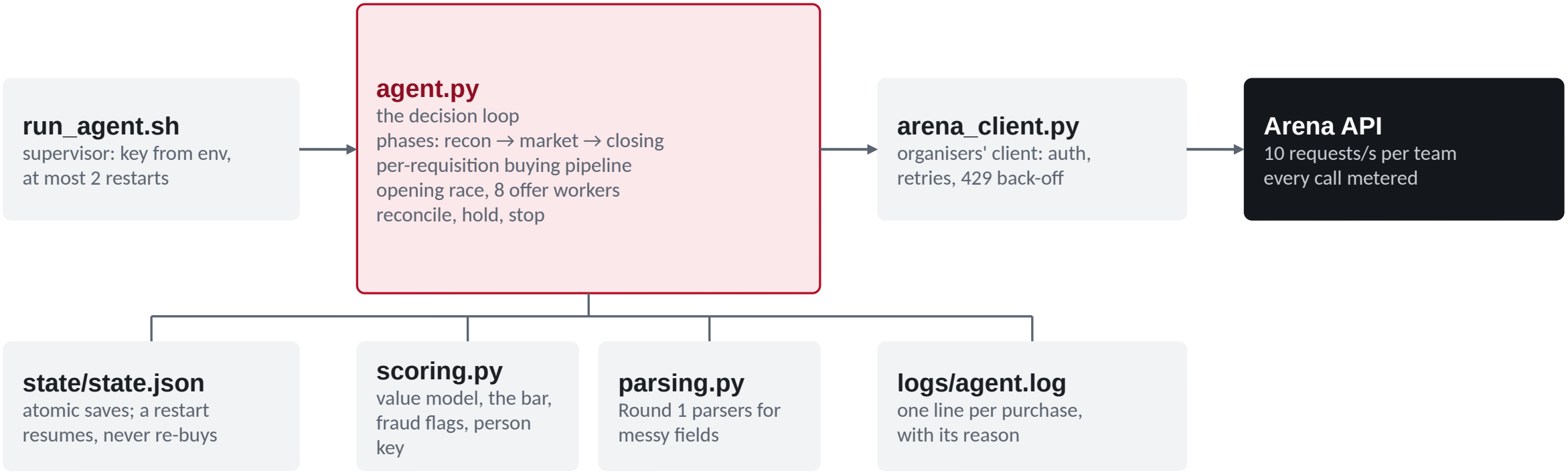
**12:23** — 10-swap pilot tests a points model

**12:30** — Ledger rejects the model; swaps off

**15:38** — Closing hour: hold, spend nothing

**17:01** — Whistle: 1,491.2, #2 of 51

# Architecture and tech stack




## Tech stack

**Python, standard library only**
No third-party packages to install or break

**ThreadPoolExecutor, 8 workers**
≈ 6 requests/s, inside the 10/s limit

**JSON state, atomic writes**
tmp file + os.replace, saved on SIGTERM

**Regex parsers, no LLM**
Messy CTC, notice and scores; keyword notes

**Bash supervisor**
Mirrors the 2-restart rule; runs 6 h unattended

# What the agent buys, in what order

STEP 1

## 12,000

**summaries read**

10 roles × 12 pages. Summaries show skills only.

1 cr / 100

STEP 2

## 3 of 6

**skills to go further**

Ranked by skill match. Nobody below 3 of the 6 wanted skills is profiled.

free

STEP 3

## 650

**full profiles**

50 per batch at 1.2 cr each; singles only when under 31 remain.

60 cr / 50

STEP 4

## 4 checks

**the bar, locally**

Role, assessment, notice, CTC. Fraud flags lower a candidate's rank.

free

STEP 5

## 150

**verified, of 192 checked**

Identity, reference and verified score, before any offer.

25 cr each

STEP 6

## 116

**signed at the opening**

Best value first across all 10 requisitions, in 10 seconds.

10 cr each

### Order of work, per requisition, every round

Verify a finalist, else profile the best unread summaries, else read another search page. Stop at 1.25 verified candidates per open slot.

### Budget guard-rails

Recon spent 6,475 credits against a hard cap of 12,000. The queues were full at 11:29, before offers opened at 11:42.

# Six decisions, and the thinking behind each

## Spend in recon, not later

Nobody can sign anyone in recon, so information bought then cannot go stale or be taken. What the agent knows at the opening decides the race.

*6,475 of our 9,876 credits went here.*

## Search wide, profile narrow

A page of 100 summaries costs 0.05 points. Profiles cost far more per person, so only strong skill matches are profiled, 50 at a time.

*1.2 credits per profile instead of 2.*

## Verify every signing

A check costs 1.25 points; an unverified signing loses 22% × 17 ≈ 3.7 points on average. The rule re-tests itself on the live failure rate.

*116 of 116 signings verified before the offer.*

## No LLM calls

/reason is metered and slow per candidate. Notes were templated, so Round 1 keyword scoring read them for free, and /assess settles zero vs points.

*0 credits spent on /reason.*

## Fire everything at the opening

The best names go first. One queue, ranked across all 10 requisitions; 8 workers use the burst; a slot and a person are reserved before each call.

*115 signed in 10 s; 11 names lost, all backfilled.*

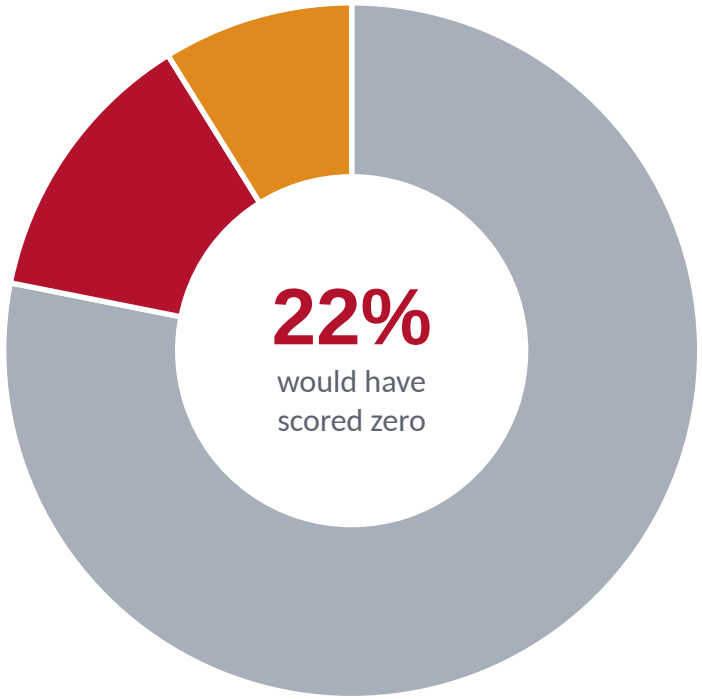
## Stop when full

Once every slot is filled, each further buy must beat its own cost. Swaps were tested once, did not pay, and were switched off.

*0 credits spent after 12:30.*

# 22% of eligible-looking profiles would have scored zero

## 192 candidates checked with /assess during recon




**150 · 78%**
passed: identity, reference and score all verified

**25 · 13%**
fabricated identity (identity_not_verified)

**17 · 9%**
real person, verified score below the bar

## Why 25 credits a check pays

**Cost of one check**
25 × 0.05 = 1.25 points

**Expected loss without it**
22% × 17 points ≈ 3.7 points per signing

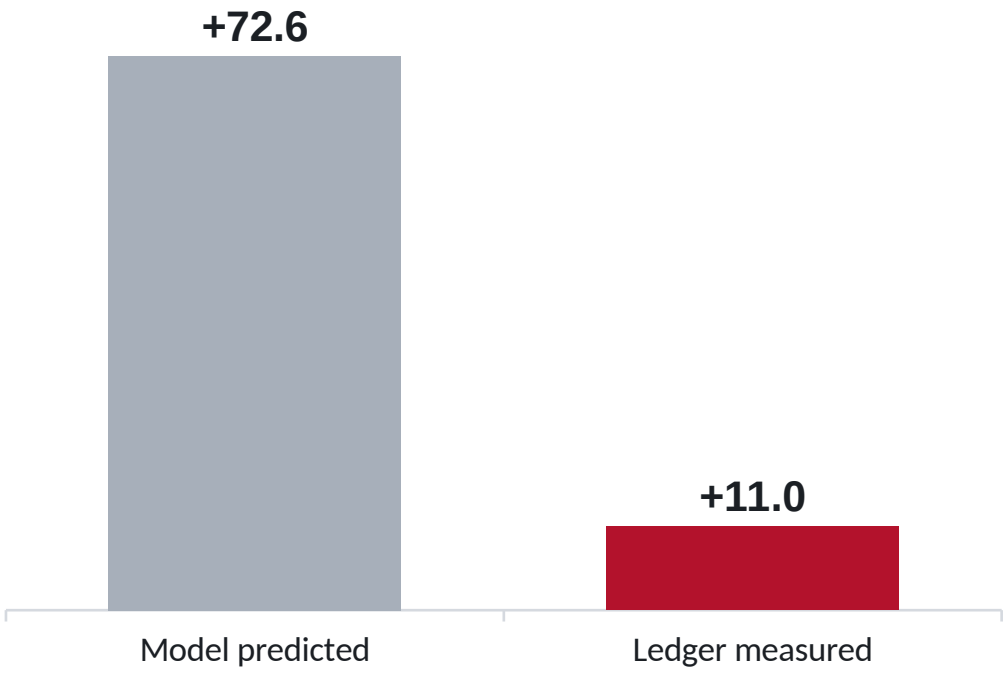
**Rule in the code**
keep verifying while fail rate × (slot value + offer cost) > check cost

**Free checks first**
stacked praise clauses (a Round 1 fraud signal), a CTC jump over 3×, impossible values; duplicates matched on name + city + experience

**Result:** 17.1 points per signing, against about 13 for teams that signed at face value.

# The experiment the agent stopped itself

## Points gained by the 10-swap pilot




1 **Hypothesis**
Points ≈ verified assessment margin. It explained 95% of our first 1,974 points.

2 **A capped pilot**
At most 10 swaps. Each replacement re-checked (2 credits) before anyone was released; if the new offer failed, the old signing was taken back.

3 **The ledger's verdict**
Accept the model only if measured ÷ predicted lands between 0.6 and 2.0. It came back 0.15, so the agent rejected its own model and swaps were switched off.

4 **Cost and lesson**
About 95 points net (2,131 credits), bounded by the cap. The two ledger readings fit points ≈ 4.19 × (skill + 2.5 × margin + notes + notice + salary).

**Takeaway:** a points model is a hypothesis until the ledger agrees. Cap the bet while you test it.

09 // EDGE CASES

# When things go wrong, the agent keeps its holdings

**Rate limit (429)**
The client backs off; a call that still fails is deferred and retried later, never counted as a failure.

**Crash or redeploy**
State saved atomically and on SIGTERM; at most 2 restarts; a restart resumes without re-buying.

**Empty search, or a role read in full**
The role is marked exhausted and weaker skill matches become eligible to profile.

**Late opening or a pause**
Never exits on 'closed'. A free phase check every second; it raced within a second of the real opening.

**404 or a malformed record**
The candidate is marked dead and never paid for again.

**Arena reset**
reconcile() compares our holdings with the server's per-requisition counts and re-signs what was lost.

**Offer rejected**
already_signed: next in the queue. requisition_full: closed locally. Same person, another ID: skipped before paying.

**Closing hour**
Offers cost 20 credits, so no swaps; every holding is kept to the whistle.

**Credits exhausted**
Every paid call stops; holdings are kept; only free calls continue.

**Stale backup candidate**
A 2-credit profile check before any release; if the new offer fails, the old signing is taken back.

**Seen live on the day:** offers opened 12 minutes late · the arena paused once · 11 names lost to rivals in the race · 0 crashes · 0 requisitions overfilled

# What we learned, and what we'd do with two more hours

## Lessons

1 **Measure before you spend.**
A live 22% failure rate made verification an easy call.

2 **Buy information while it cannot go stale.**
Recon was our cheapest hour, and it decided the race.

3 **A model is a hypothesis until the ledger agrees.**
A capped pilot limited a wrong guess to about 95 points.

## With two more hours



**Rank on all five points terms from the start**
and profile far more candidates: the leader's 21.5 points per signing shows the headroom.



**Measure the points formula early**
with small, deliberate swaps, not one late pilot.



**Batch-refresh stale backups**
(60 credits per 50) before any release.

# Thank you. Questions welcome.

AI assistant: Claude was used for help in data analysis, writing and testing the agent's code. The team reviewed and approved every deployment and decision.